



# Vaishnavi Goyal

Student at PES University

## About Me

Computer Science undergraduate at PES University with a focus on Quality Engineering and Adversarial Machine Learning. Experienced in building automated testing frameworks, container runtimes, and low-level system monitoring tools. Passionate about applying rigorous testing techniques and security-focused automation to ensure the resilience and quality of large-scale financial software systems.

## Contact

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 <https://portfolio-l1bv.onrender.com/>

## Education and training

### (2021-2022)

SSC, CBSE board  
95.4%

### (2023-2024)

HSC, CBSE board  
90.6%

### (2024-2028)

PES University  
B.Tech  
Computer Science  
CGPA : 8.42  
Scholarship Recipient



## Skills

### Technical Skills:

- Programming : Python (Primary), C/C++, JavaScript (Node.js), Java.
- Data & Analytics : Pandas, NumPy, Matplotlib, SciPy, Statsmodels.
- Testing & Automation : Automated Test Frameworks, White-Box Testing, Stress Testing, Regression Analysis, Statistical Testing.
- Systems & Infrastructure : Linux Kernel Modules, Container Runtimes (Docker-like), UNIX Sockets, Multithreading/Synchronization, Networking, DBMS.
- Concepts : Adversarial Machine Learning (AML), Lexical Analysis, Context-Free Grammar (CFG), Data Structures & Algorithms.
- Web & tools : HTML5, CSS, React, Node.js, Figma, Canva.
- Security & AML: Adversarial Stress Testing, Red Team Simulation, Vulnerability Assessment, Security Lifecycle Management.



## Projects :

### 1. **Adversarial Machine Learning (AML) Security Framework**

- Engineered an end-to-end security lifecycle engine for adversarial stress testing on financial AI models (XGBoost, DistilBERT, ResNet), implementing white-box attack vectors (FGSM, PGD) to identify vulnerabilities in high-stakes transaction logic.
- Architected an automated defensive pipeline integrating adversarial training and input sanitization, which proactively hardens models against adversarial drift and maintains data integrity.
- Built a centralized telemetry system using SQLAlchemy/SQLite for structured logging of adversarial samples, detection outcomes, and performance metrics, creating an immutable audit trail for security compliance.

### 2. **Lightweight Container Runtime & Kernel Monitor**

- Designed and implemented a mini-Docker runtime in C, utilizing UNIX socket IPC and Linux kernel modules to monitor memory and process synchronization—demonstrating low-level systems expertise and OS-level debugging capabilities.

### 3. **JavaScript Syntax Validator (Compiler Design)**

- Developed a robust validation tool in Node.js utilizing lexical analysis, recursive descent parsing, and Context-Free Grammar (CFG), showcasing the ability to build sophisticated automation and validation tools from scratch.

### 4. **House Sales Data Analysis**

- Performed rigorous Exploratory Data Analysis (EDA) and predictive regression analysis on large-scale datasets, establishing a data-driven quality assessment framework to validate model accuracy and predict price variances.

### 5. **Intelligent Automated Waste Segregation System**

- Engineered an embedded real-time classification system using sensor fusion (ultrasonic, IR, moisture) and servo-control logic, demonstrating hardware-software integration and real-time decision-making optimization.

### 6. **Full-Stack & Auction Systems (E-commerce Architecture)**

- Designed scalable web-based platforms incorporating secure user authentication, transactional bidding logic, and responsive UI interfaces, highlighting proficiency in full-stack integration and end-user performance optimization.

### 7. **Advanced Port Scanner**

- Developed a custom network auditing tool for service detection, applying deep knowledge of network protocols and socket-level communication to identify infrastructure exposure and security risks.